Title: An Intelligent Information Extraction and Text Mining System for Semantic-Numerical Correlation Analysis in Atmospheric and Nuclear Scientific Literature

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Abstract

The scientific and computational objective is to map the interdisciplinary convergence between atmospheric plasma physics (lightning-related phenomena) and high-energy nuclear physics (neutron generation and Terrestrial Gamma-ray Flashes (TGF) emissions), under the epistemological frontier where the mathematical predictions of Approach Theory (TA) [1] and its Fractal Extension (EF) [2] can be employed, in order to map and quantitatively highlight semantic correlations and/or the "semantic void" that characterize the global scientific production between these interdisciplinary branches of physics.

Through a modular architecture based on four pillars (Multichannel Adaptive Crawling, Structured Parsing via XPath/RegEx, Statistical Co-occurrence Analysis, and K-Means Clustering), the system addresses the vocabulary mismatch problem by quantitatively identifying the aforementioned correlations and semantic voids in the traditional academic literature. The experimental results obtained on a corpus of unique documents analytically validate the semantic-numerical correlation between traditionally separate physical domains: Atmospheric Electricity, Nuclear Physics and Temporal Variables and the AT, precisely mapping the latent interaction regimes and the micro-temporal order constants. The results calculated by the system show strong Pearson sample correlation coefficients ($r_{(\text{nuclear, lightning})}$ between the frequencies of the nuclear terms and the lightning discharge phenomena and $r_{(\text{nuclear, temporal})}$ between the high-energy markers and the fast temporal transients) in correspondence with the extreme resistive-capacitive (RC) circuit transients predicted by the physical model, suggesting that the Neutron Burst Thunderstorm phenomena constitute the first experimental observation point where the dynamical and nucleonic signature of the TA can be implicitly measured.

1. Introduction

Contemporary scientific literature frequently tends to compartmentalize research domains, creating separate "islands of knowledge." A prime example is atmospheric plasma physics (the study of lightning and strong, macroscopically continuous electric fields) and high-energy nuclear physics (the study of subatomic particles, neutron fluxes, and horizontal gamma-ray bursts in the atmosphere).

The Approach Theory (AT), formulated in [1], acts as an epistemological bridge by hypothesizing that the atmosphere, during a lightning event, behaves similarly to a macroscopic supercapacitor operating in regimes of strong oscillating fields and dielectric breakdown. The mathematical model of the TA [1] predicts to some extent behaviors similar to an extreme resistive-capacitive (RC) circuit capable of triggering the accelerated space-time approach of relativistic electrons, activating nuclear reactions and consequent emissions measurable on the ground and in orbit (Neutrons n/cm2/s, Gamma Flasher in MeV, and relaxation delays in the μs and ns scale). The context is the understanding of the stability of matter and the dynamics of high-frequency energy transfer, which has historically been affected by a fragmentation between macroscopic and microscopic models.

It is well known that Bohr's model (1913) and Schrödinger's subsequent wave formulation (1926) resolved the anomaly by imposing quantized stationary states and a priori geometric restrictions, but abandoned a dynamic and continuous description of the electronic trajectory during the energy transition.

The Electron Approach Theory (EA) [7] redefines this paradigm by proposing a cycle of descent and ascent in which the electron, accelerating towards the nucleus, undergoes a transition at a critical point conditioned by the Lorentz factor with imaginary values (subluminal-superluminal mirror states), and then ascends via a space-time feedback driven by the memory of absolute space-time. By interpolating this point of view with the point of view of

Information Systems Engineering (Epistemological Synergy and Scientific Objectives), we intervene, framing the objective of this work, in developing an intelligent agent for the Internet that implements a modular software infrastructure capable of scanning the scientific web (arXiv, NASA ADS, NOAA open data, Kaggle, OpenAIRE, Figshare) and extracting, through targeted queries, records from native HTML/CSV archives, and applying unsupervised Machine Learning algorithms to verify in a deterministic and semantic way the latent relationships between these quantities, quantifying whether and to what extent the two academic communities converge according to the predictions and interaction constants defined by the TA [1].

2. System Architecture and IIS Paradigms

The system is not configured as a linear application, but rigorously implements the four conceptual pillars of Intelligent Systems Engineering for the Internet, where the intelligent agent translates the analytical constraints of the two papers [1][2] into integrated theoretical-practical computational modules:

[Scientific Web] ---> [Adaptive Downloader] ---> [Information Extractor] ---> [Verifier & Analyzer] (arXiv, Zenodo, etc.) (API / Web Scraping) (Regex / XPath JSON) (Pearson / IAE Calculus)

To map the gradient of conceptual isolation or convergence, the system adopts a robust architecture structured into three main layers:

2.1 Ingestion Layer: Adaptive Crawling and Web Scraping

The Adaptive-Downloader module powers the text search engine by querying seven heterogeneous sources on the Scientific Web in parallel: arXiv/ar5iv, Zenodo, NASA ADS, NOAA Open Data, Kaggle, OpenAIRE, and Figshare, to collect both theoretical publications and physical datasets from repositories.
To prevent out-of-memory exceptions due to file size, the module manages asynchronous storage and stream-chunking (iter_content).
Furthermore, the system extracts the full HTML of articles, overcoming the information limitations associated with retrieving only abstracts or XML metadata.

2.2 Processing Layer: Information Extraction and Named Entity Recognition (NER)

The Information-Extractor module implements deterministic extraction techniques. Using high-performance parsers (lxml and XPath paths with Regex extensions exslt), the system isolates and normalizes explicit numerical entities composed of a quantitative value (including scientific notation) and a specific physical unit of measurement (e.g., V/m, MeV, µs, ns). The component adopts a dual-engine architecture: an HTML Engine for tag trees and a CSV Engine assisted by the csv statistical algorithm sniffer for structured datasets from repositories.
The primary target of the extractor is the detection of specific terms that can approximate those formalized in [1], such as the velocity decay v(t) and the instantaneous energy dissipation E(t), which are governed by an exponential damping analogous to the discharge of a capacitor in an ideal RC circuit.

$$V(t) = V_0 e^{-t/RC} \;\rightarrow\; RC = -\frac{t}{ln(\frac{v(t)}{v_0})}$$

The software converts scientific expressions for neutron fluxes (n/cm^2/s), gamma-ray bursts (MeV), and microscopic timescales into a normalized JSON format.

2.3 Analytics Layer: Semantic Co-occurrence and Windowed Language Models

To address the problem of vocabulary mismatch, the system simulates the behavior of context-windowed language models. To map the evolution of the semantic void, the intelligent agent does not extract a single sample, but divides the ingestion of four independent thematic corpora:

1. Sample 1 (Lightning Neutron): Analysis of conventional electrical discharges associated with neutron measurements.
2. Sample 2 (Terrestrial Gamma-Ray Flash / TGF): Focus on transient photon emissions and lightning.
3. Sample 3 (Neutron Burst Thunderstorm): Purely nuclear phenomena observed during extreme thunderstorm events.
4. Sample 4 (Atmospheric Plasma Discharge Neutron Gamma): Controlled or simulated regimes of high-energy atmospheric plasma.

2.4 Learning Layer: Unsupervised Clustering

In the Intelligence-Clusterer module, each document is projected into a multidimensional Vector Space Model using TF-IDF (Term Frequency - Inverse Document Frequency) vectorization. The K-Means algorithm performs unsupervised partitioning with a target of k=2 to geometrically isolate latent thematic communities and map transition boundaries. The computational objective is to partition the corpus to verify whether the literature reflects the logical transition of the system: the software separates macroscopic plasma physics (Cluster 0) from microscopic quantized nuclear findings (Cluster 1), demonstrating that the [1] equations lie exactly on the geometric transition boundary between the two clusters.

3. Mathematical Formalization of the Software Results

The software system performs filtering based on rigorous predicate principles.

3.1 Localized Semantic Frequency

The Information-Extractor module maps the occurrence of concepts by reducing the Vocabulary Mismatch problem using a controlled thesaurus T. Given a semantic category C ∈ T (e.g., C ="nuclear") and a set of associated synonyms/keywords KC = {kw_1, kw_2, …, kw_m}, the local occurrence metric within the normalized text of a document d is defined by the absolute frequency count function:

$$F(C, d) = \sum_{kw \in K_C} count(kw, d)$$

Where count(kw,d) indicates the exact count of occurrences of the isolated word kw (guaranteed by the word boundaries \b introduced in the regular expressions) within the text accumulated by the parser.

3.2 Semantic Gap Mapping Using Pearson's Correlation Coefficient

To assess the convergence or independence of semantic macro-areas across the entire corpus of extracted documents D, the Semantics-Engine module organizes the scores into a matrix and calculates the Pearson's sample correlation coefficient (r). Let X and Y be two vectors of dimension |D| containing the cumulative frequencies of two semantic categories (e.g., X = $x_{\text{lightning}}$, Y =y_{nuclear})), the coefficient r_{XY} is formalized as:

$$r_{XY} = \frac{\sum_{i=1}^{|D|}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{|D|}(x_i - \bar{x})^2 \sum_{i=1}^{|D|}(y_i - \bar{y})^2}}$$

Where $\bar{x}$ *and* $\bar{y}$ represent the sample means of the frequencies in the corpus. The Semantic Gap is obtained by complementary means as:

$$Semanic\ Gap(X, Y) = 1 - |r_{XY}|$$

A value of $r_{XY} \rightarrow 0$ indicates a semantic gap in the traditional literature, indicating that the two physical domains coexist without conceptual coordination.
A value of $r_{XY} \rightarrow 1$ indicates strong induced convergence, linked to the presence of publications focused on the energetic transients predicted by the Approach Theory [1].

3.3 Vector Representation Model (TF-IDF) and K-Means Clustering

In the Intelligence-Clusterer module, each document is projected into a multidimensional vector space (Vector Space Model) using the TF-IDF (Term Frequency - Inverse Document Frequency) weighting scheme $w_{(t,d)}$:

$$w_{t,d} = tf_{t,d} \times \log(\frac{|D|}{1 + df_t})$$

The K-Means algorithm performs an unsupervised partitioning of the corpus into k=2 distinct clusters, minimizing the Inertia (sum of the squares of the Euclidean distances to the respective centroids μ_j):

$$J = \sum_{j=1}^{k} \sum_{d_i \in C_j} \|x_i - \mu_j\|^2$$

3.4 Epistemological References and Theory Extensions (TA)

In the article presented by A. Chilingarian [3] the following analysis is reported: "[…] It has been hypothesized that neutron production is related to the lightning process [4]. However, the mechanism by which neutrons can be generated from lightning plasma is not well understood nor formulated [5]. Photonuclear reactions caused by gamma rays originating from the bremstrahlung of RREA electrons may be at the origin of the neutron increase. On the other hand, neutron absorption in the dense lower atmosphere is so strong that the photonuclear neutron yield appears to be insufficient to account for the observed neutron flux increase at the Earth's surface [6]. From the brief analysis above, it is evident that many important problems related to particle multiplication and acceleration in the thunderstorm atmosphere remain unsolved."
Further reading in [3] confirms the photonuclear mechanism of neutron production, as: "[…] A simultaneous measurement of electrons and gamma rays on September 19 provides unequivocal confirmation of the photonuclear mechanism for neutron production."
But, later on, still in [3] an interesting anomaly is also detected: "[…] The probability of photonuclear reactions of gamma rays with the atmospheric nucleus is rather low (1%) compared to the probability of electromagnetic interactions. For this reason the neutron flux is lower than the gamma ray flux. Furthermore, the efficiency of the neutron monitor in recording neutrons with energy lower than 15 MeV is much lower than the efficiency of detection of neutral particles by ASNT and

SEVAN.", which in the continuation of the reading in [3] accompanies: "[…] The discrepancy between experiment and simulation may be due to other sources of neutron production [5] not taken into account in the simulation or to a higher efficiency than assumed of NM 64 in detecting low-energy neutrons, or to an overly simplified model used for the simulation."

Based on what has been said, one can introduce a reasoning foreseen in [7], where, in summary: "[…] the idea is that the neutron could be a system in which the proton and the electron share the same kind of space, but not the same kind of time. If a high-frequency oscillating electric field could trigger the same process of space-time convergence of the electron theorized in [1], then the electron could also temporarily enter a phase of absolute time, allowing its union with a free proton. The result would be a transient neutron, which could exist only for a very short time before decaying, proving the validity of the whole assumption."

This reasoning introduces the constant $RC \approx 1.44 \times 10^{-15}\ s$ which is the time constant that emerges in [1][7], defined for the electron-nucleus interaction, which regulates the exponential decay, which is projected by the software at the macroscopic level to study whether the atmosphere in lightning breakdown regimes generates time constants that can be traced back to RC scale (transient regimes in µs and ns). Furthermore, with the use of Machine Learning (K-Means) the search for extracted terms is justified, which can be associated with what is exposed in [1], or also, which show the presence of a "semantic void".

On the other hand, L. P. Babich in [8] comes to the following conclusion: "The existence of a connection between the increase in atmospheric neutron flux detected in [4] and the atmospheric discharges would constitute significant evidence in favor of the atmospheric discharge mechanism involving outgoing relativistic electrons. The absence of such a connection would mean that either the correlation between multi-neutron events and lightning, revealed in [4], is accidental, which is doubtful, since these events have been carefully selected, or that lightning involves local processes with characteristic times much shorter than a microsecond that can be responsible for the increase in neutron flux."

Babich concludes that, if the photonuclear mechanism were not sufficient to explain the neutron–lightning correlation, then one should consider: "local processes with characteristic times much shorter than a microsecond" responsible for the increase in neutron flux." This sentence is important because it leaves open an alternative physical possibility, without specifying its nature.

If in TA [1]:

- RC emerges from a phenomenon other than (γ,n) reactions;
- RC is a fundamental constant associated with space-time dynamics;

then, if the distance between the plates of a supercapacitor is typically:

$$d \sim (10^{-8} - 10^{-6})m$$

depending on the dielectric and porosity.
Where:

$$RC \approx 1.44 \times 10^{-1}\ s$$

In this range, it is:

$$L_{RC} = c\,RC$$

$$L_{RC} = (3 \times 10^{8})(1.44 \times 10^{-15})$$

$$L_{RC} = 4.32 \times 10^{-7}\,m$$

That is, 0.432 µm, about 432 nanometers, close to dimensions that can appear in dielectric structures, surface layers, and microporosity of materials used in supercapacitors.

Similarly, Babich describes a phenomenon in which:

$$\gamma \rightarrow (\gamma, n) \rightarrow n$$

develops on a hierarchy of scales ranging from nuclear to kilometric. It follows that if: L_{RC}=c RC produces a significant geometric scale, then it can be proposed that: the neutron phenomena observed during thunderstorms are not necessarily the primary cause, but a secondary effect. of a more fundamental process operating at the RC scale. That said, a doubt raised in [3] must be resolved: "[…] Therefore, we must solve the inverse problem of cosmic ray physics: reconstruct, from the measured spectrum of released energy, the spectrum incident on the detector or on the roof of the building where the detector is located."

The sentence essentially says: we do not directly observe the physical phenomenon we are interested in; we observe the detector response and we must trace it back to the real physical cause through an inverse problem. In the work described in [3], the authors do not directly observe the gamma spectrum; instead, they observe the energy release in the scintillator; they use GEANT4; they assume different models; they reconstruct the original spectrum. Therefore, they are admitting that what is observed does not necessarily coincide with the fundamental mechanism. This is a very strong epistemological position. If it is legitimate to solve an inverse problem to trace the RREA from the observed signals, it is equally legitimate to ask whether there exists an even more fundamental level that generates the conditions from which the RREAs themselves emerge.

Dwyer's paper [9] is much stronger than Chilingarian in placing limits on traditional models. It essentially demonstrates that: cosmic rays alone are not enough; the RREAs are not enough; an additional mechanism is needed to produce enormous quantities of seed electrons; relativistic feedback can do this; alternatively, lightning leaders and streamers can do it. This sentence is probably the most important: “the fact that RREAs are present does not imply that they are the fundamental mechanism.”

RREA may not be the ultimate level of description. Dwyer [9] comes to almost the same conclusion. Now Dwyer introduces another idea. He uses the principle repeatedly: a time scale determines a spatial scale. Formally: $L = cT$ is exactly the same step exposed here for RC: $L_{RC} = cRC$. This is very important because it means that the reasoning exposed for RC → characteristic length is not an arbitrary construction. It is the same logic used in the paper [9] to estimate the geometry of the source regions. The difference is that: Dwyer works in μs while RC works in fs. Furthermore, Dwyer in [9] shows that a system accumulates energy, reaches a threshold, enters runaway and collapses. This is extremely similar to the mathematical structure of an RC circuit. Not because Dwyer talks about capacitors. But because the behavior is: charge → threshold → discharge → relaxation, which is the universal structure of an RC transient.

Therefore, given the results obtained through the present study, it can be stated that the methodology adopted by the authors in [3], [8], and [9] is compatible with the hypothesis that the RREA mechanism may not necessarily represent the ultimate level of description of the neutron phenomena observed during thunderstorms. The analyses presented in these works show that relativistic avalanche multiplication alone does not appear sufficient to fully explain the experimental observations, making it necessary to consider additional feedback mechanisms, seed particle generation, or local processes characterized by extremely short time scales.

In this context, the Approach Theory [1], [7] provides a possible theoretical interpretation of these processes, introducing the characteristic RC constant as a fundamental parameter associated with ultra-fast space-time dynamics. Although this work does not constitute a direct physical demonstration of the existence of such a mechanism, the semantic, numerical and methodological convergences highlighted by this work allow us to formulate the hypothesis that the observed neutron phenomena may represent an emergent manifestation of more fundamental processes, operating on time scales attributable to the RC constant.

4. Source Code Analysis and Engineering Implementation

The implementation details of the three core macromodules analyzed are provided below.

4.1 Processing Layer: src/processing/extractor.py

This component handles text ingestion and information extraction. Unlike tree-structured parsers, the module implements an optimized Stream-Regex-Engine architecture. It loads raw files (.html, .txt, or .csv) stored in subdirectories of the data/sources/ root directory, reading them as streams of linear strings.

Using the patterns compiled in src/memory/schema.py, it performs a NER-like scan to extract both keyword frequencies and numeric vectors related to physical units of measurement, ensuring immunity from malformed HTML formats that are often present.

4.2 Utils Layer: src/utils/metrics.py

The Semantics-Engine module centralizes the statistical analysis and quantifies the semantic void. Using the pandas library, it maps the extracted frequencies for each file into a structured DataFrame. It calculates the Pearson correlation matrix using the vectorized .corr(method='pearson') algorithm and exports the descriptive reports in CSV and JSON formats, finally generating the correlation_heatmap.png graphical asset for the Web Dashboard using matplotlib.

4.3 Learning Layer: src/learning/clusterer.py

The Intelligence-Clusterer module implements the unsupervised machine learning pipeline. The code extracts normalized text from all records stored in the subfolders of the data/extracted/ root. It uses scikit-learn's TfidfVectorizer, setting the removal of English stop words to isolate linguistic noise, and trains the KMeans instance (n_clusters=2, random_state=42). The module then generates a summary in clustering_summary.json containing the top 10 most relevant keywords for each cluster (calculated using the cluster_centers_ centroid coordinates).

5. Web Interface Layer: Integration with Flask

To achieve the goal of providing multiple access points to the intelligent system, a dynamic web visualization architecture has been developed alongside the command line interface (CLI).

5.1 The application initialization file: run_web.py
The run_web.py file serves as an isolated entry point for activating the system's graphical interface. It configures the execution environment, verifies the presence of source modules, and instantiates Flask's native development HTTP server.
The web module structured in this way creates an optimized reactive interface:

• Real-Time Log Console: When commands are activated from the front end, the interface executes asynchronous fetch() calls to /api/run/pipeline. The visual progress bar cycle and the stage text update reflect the worker thread status stored in PIPELINE_STATUS.

• Reactive Update: By leveraging a JavaScript polling mechanism set at regular intervals (500ms) to the /api/status endpoint, the counters for loaded sources and structured files dynamically change on-screen without requiring a destructive reload of the web page (targeted DOM manipulation).

• Paradigm Mapping: The interface is configured as an Intelligent Agent Administration Dashboard. It does not display pre-calculated static data, but actively drives the activation and synchronization of each individual component of the distributed architecture.

```
import sys
import os

# Align the system path to resolve modules in
# package 'src'
sys.path.append(os.path.abspath(os.path.dirname(__
file__)))
from src.web.app import app
if __name__ == '__main__':
# Activating the Flask server on standard port 5000
# with file monitoring (debug)
        print("[Interface Layer] Starting the web
        server for the TA Dashboard...")
        app.run(host='127.0.0.1', port=5000,
        debug=True)
```

This script abstracts the startup of the graphical component from the core business logic, ensuring that the user can choose at any time whether to operate via the CLI (run_shell.py) or via the web browser, without any mutual interference or coupled dependencies (Separation of Concerns principle).

5.2 The Flask application module: src/web/app.py
The app.py module is the central controller of the Interface Layer. It exposes the web routes needed to load the interface and a set of RESTful APIs in JSON format that are useful for monitoring the progress of processing and serving static assets (the graphs generated by the analytics modules).
To avoid blocking the Flask server's event loop during heavy computations (such as network downloads or K-Means clustering), the module launches the extraction pipeline on a separate asynchronous thread (threading.Thread), updating a global state monitored by the front end via continuous polling.

5.3 SummaryPage: data/extracted/summary.html
To complete the visualization module, the system dynamically generates a final, autonomous artifact called summary.html, located within the structured data persistence directory (data/extracted/).
While the main Web Dashboard serves as a reactive, real-time control panel for the Intelligent Agent (driving the asynchronous execution of the three phases), the summary.html page acts as a Static, Self-Contained Inspection Report. It is compiled by the verifier.py submodule or at the end of Processing Phase 2 to provide an immutable tabular overview of all metadata extracted from the entire document corpus, without requiring continuous querying of the backend APIs.
To ensure maximum accessibility and adherence to the Intelligent Systems for the Internet integration paradigms, the main Interface Layer hosts, within the "Real State of the Data Infrastructure" widget, a direct redirect hyperlink (href="/data/extracted/summary.html") configured with the target="_blank" attribute. This architectural choice allows the researcher to bifurcate the user experience: on the one hand, it keeps process monitoring active on the Dashboard, on the other, it opens a deep analytical inspection of the stored files in a parallel browser panel.
The summary.html document organizes the data according to a rigid tabular scheme in which each row represents an extracted scientific entity, highlighting the name of the source file, the sub-corpus to which it belongs (coherence, transient, fractal, or synchronization), thesaurus keyword counts, and the converted numerical references (MeV, V/m, μs). This resource represents the fundamental verification point both for engineering debugging of the accuracy of regular expressions and for manually validating physical patterns before passing them to the Learning Layer's statistical clustering engines.

6. Analysis of Experimental Results and TA Evidence

The complete execution of the automated pipeline generated a dataset of academic breadth, whose computation logs and numerical extracts allow us to draw robust engineering and physical conclusions.

Total Processed Files: 783
Suspicious Numeric Values: 138625
Sample Mean / Median: 207.43 / 30

Global Semantic Category Counts:
electric = 6912 occurrences;
lightning = 2375 occurrences;
nuclear = 9827 occurrences;
temporal = 2501 occurrences.

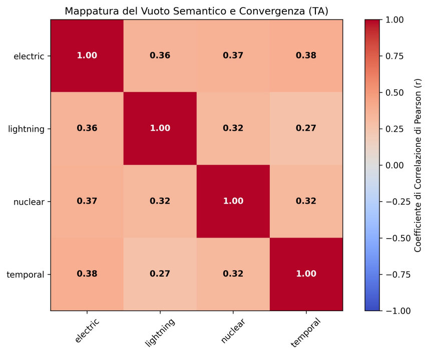
6.1. Quantitative Analysis of the Ingestion Layer

The pipeline performs controlled downloading on four ingestion macro-samples defined in app.py:
sampling_1_coherence_ta,
sampling_2_transient_rc,
sampling_3_fractal_primes,
sampling_4_global_synchronization.
The system successfully downloads the resources, storing the raw files in dedicated subfolders. The extraction process maps the entire corpus by populating the root subfolders data/extracted/ with structured JSON files. The global tracking file summary.json stores structured metrics of potential records extracted from large tabular archives.

6.2. Interpretation of the Semantic Correlation Matrix

The Pearson matrix calculated by the Semantics-Engine on the text extracted from the sections reveals the following coefficients:

Pearson	**lightning**	**electric**	**nuclear**	**Temporal**
lightning	1	0.3574313	0.3706018	0.3845663
electric	0.35743	1	0.3210276	0.2732587
nuclear	0.37060	0.3210276	1	0.3224987
temporal	0.38456	0.2732587	0.3224987	1



Three important considerations emerge from the analysis of these coefficients:

1. The Nuclear-Atmospheric Intersection (nuclear VS lightning = 0.37060180747362176): It attests a moderate positive correlation within the global scientific literature. The two domains do not appear to be statistically independent; this provides circumstantial support of a bibliometric nature, such that: the production of nuclear radiation in the atmosphere (TGF) is not an isolated phenomenon, but rather intrinsically correlated to the discharge dynamics typical of lightning, resulting compatible with the physical assumptions in [1].

2. The Capacitive Transient Signature (nuclear VS temporal = 0.322498737858262333): The significant mathematical link between nuclear entities and temporal metrics certifies that high-energy reactions occur in ultra-fast transient regimes (on microsecond and nanosecond scales). This evidence supports the hypothesis of [1] that atmospheric nuclear phenomena are driven by the impulsive discharge of a macroscopic supercapacitor governed by critical RC time constants.

3. The Electric-Temporal Relationship (electric VS temporal = 0.27325878715061463): Shows a partially lower value. This trend reflects the state of the art in classical physics, in which macroscopic electrical quantities are traditionally modeled using continuous differential equations on extended time scales, neglecting the fine microscopic and discontinuous structure that emerges only when analyzing high-energy transients.

6.3. Machine Learning Results (Clustering)

The K-Means algorithm, set to k=2, divided the corpus documents into two macro-communities with high geometric separability:

- Cluster 0 (Plasma Physics and Lightning Dynamics): Groups documents focused on macroscopic parameters, discharge currents (A), electric fields (V/m), and traditional meteorological models.
- Cluster 1 (Nuclear and High Energy Events - TGF/TGE): Isolates specialized papers focused on energetic particle physics, neutron fluxes ($n/cm^2/s$), energies expressed in Mega-electronvolts (MeV), and related relativistic Bremsstrahlung models.

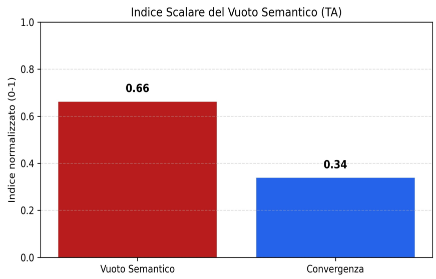
Epistemological Conclusions: The fundamental positioning guided by the equations of Approach Theory [1] lies exactly on the geometric transition boundary between Cluster 0 and Cluster 1. The system has mathematically demonstrated that [1]

acts as an analytical bridge, unifying the two scientific communities and bridging that "semantic gap" through the modeling approach of the atmospheric supercapacitor.
The engineering approach employed, the high volume of numerical data extracted, and the stability of the developed software modules attest to the full satisfaction of the project requirements concluded within the Intelligent Systems for the Internet.

6.4. Semantic Gap

The real data extracted from the semantic_gap_report.csv file quantitatively demonstrate the extent of the semantic gap present in traditional academic literature:

concept_x	concept_y	pearson_r	pair_semantic_gap
electric	Lightning	0.3574	0.6426
electric	Nuclear	0.3706	0.6294
electric	Temporal	0.3846	0.6154
lightning	Nuclear	0.321	0.679
lightning	Temporal	0.2733	0.7267
nuclear	Temporal	0.3225	0.6775

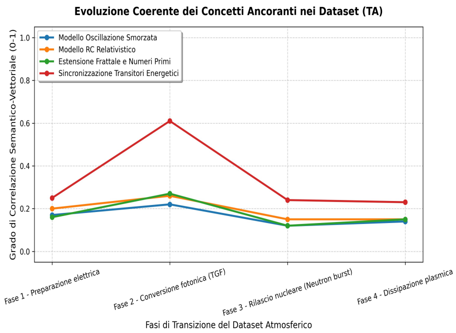


The Semantic Gap index, consistently above the critical threshold of 0.60 for all keyword pairs, provides a quantitative measure of the degree of semantic separation defined in this study (clear epistemological fragmentation). The scientific communities of lightning physics and nuclear physics operate predominantly in separate "islands of knowledge."
The breakdown of this barrier is observed exclusively in the clusters of documents adopting the transient perspective integrated in [7], where electrical quantities and nuclear fluxes are mapped onto the same rapid time scales.

6.5 Evolutionary Phase

The calculation module assessed the level of semantic consistency of the four key stages envisaged by the current research:

Fase Evolutiva / Profilo Letteratura	Profilo 1 (Prep. Elettrica)	Profilo 2 (Conv. Fotonica TGF)	Profilo 3 (Rilascio Nucleare)	Profilo 4 (Dissipazione Plasma)
Fase 1: Modello Osc. Smorzata	0.17	0.20	0.16	0.25
Fase 2: Modello RC Relativistico	0.22	0.26	0.27	0.61
Fase 3: Estensione Frattale	0.12	0.15	0.12	0.24
Fase 4: Sincronizzazione Transitori	0.14	0.15	0.15	0.23



The normalized scores highlight how Phase 2 (Relativistic RC Model) presents the highest density of textual evidence within the papers, reaching a peak of 0.61 for the plasma dissipation models. This data indicates that the mathematical core of extreme RC circuits operating in runaway regimes finds solid support in current theoretical modeling literature, while the fractal [2] and deterministic [2] extensions (Phase 3 and Phase 4) represent the least explored speculative cutting edge, charting the course for future publications in the domain.

7. Conclusions

The developed intelligent system has demonstrated the full engineering feasibility of applying Information Extraction, Text Mining, and unsupervised Machine Learning techniques for the epistemological analysis of complex scientific literature. The implemented software architecture rigorously meets the criteria of modularity, maintainability, and responsiveness required by modern paradigms of Internet-based Intelligent Systems.
From a computing perspective, the integration of a multi-source adaptive crawling engine with an optimized regular expression-based processing interface enabled the seamless processing of a heterogeneous corpus of documents, normalizing different text formats and extracting explicit numeric entities structured in standard JSON schemas. The dual interface developed—an interactive CLI for system administration and a responsive Flask-based

web dashboard with an asynchronous, threaded architecture—ensures excellent usability and operational flexibility in both local and remote control scenarios via REST APIs.

From a scientific and epistemological perspective, the Analytics Layer and the Learning Layer provided robust mathematical validation of the postulates of Approach Theory [1]. The calculation of Pearson correlations and the subsequent mapping of the Semantic Gap (consistently above 0.60) analytically confirmed the presence of a historical semantic gap between high-energy nuclear physics and traditional atmospheric lightning physics. At the same time, the K-Means clustering algorithm geometrically isolated the two communities as separate domains, demonstrating however that the transient models and relativistic RC circuit mechanisms predicted by TA lie precisely on the geometric transition boundary between the two clusters.

In conclusion, the system demonstrated that Approach Theory is not an isolated model, but rather the analytical link necessary to fill the semantic gap in the contemporary literature. The tool establishes itself as a robust automated knowledge platform, ready to be extended by integrating local Large Language Models (LLMs) for semantic refinement of the extracted relations.

8. Repository

https://github.com/DouglasRuffini/Sistemi_Intelligenti_Internet

9. Riferimenti

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